

Database Management Systems

CPS 3740

Lecture 14

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Outline

- Continue last lecture
- Create tables with cascade
 - Cascade
- ALTER and foreign key
 - INSERT, UPDATE, DELETE
- Continue Chapter 7
 - How to use the integrity enhancement feature in the ALTER TABLE statements.

Use MySQL Workbench

- We will use MySQL workbench for the complicated SQL statements.
 - Create tables with foreign keys and cascade
- But you should still know how to run them on the command line.
- The command line is very important to DBA.

IEF - Referential Integrity

- FK is column or set of columns that links **each row** in **child** table containing foreign FK to row of **parent** table containing matching **PK**.
- Referential integrity means that, if FK contains a value, that value must refer to existing row in parent table.
- ISO standard supports definition of FKs with FOREIGN KEY clause in CREATE and ALTER TABLE:
FOREIGN KEY(branchNo) REFERENCES Branch

Referential Integrity Constraints

Table: Students

Primary Key

<u>id</u>	name	zipcode
1001	Austin	7123
1002	Mary	8348
1003	Claudia	91384

Foreign Key

Primary Key

Relationship

<u>id</u>	name	<u>s_id</u>
1	Java	1001
1	Java	1002
2	Database	1001

Table: Courses

- Insert data: which record/key should exist first?
- Delete data: which record/key should be deleted first?

Textbook examples in MySQL

```
create table Hotel(  
    hotelno varchar(10),  
    hotelname varchar(20),  
    city varchar(20),  
    primary key (hotelno) );
```

```
create table Room(  
    roomno numeric(5),  
    hotelno varchar(10),  
    type varchar(10),  
    price decimal(5,2),  
    primary key (roomno, hotelno),  
    foreign key (hotelno) REFERENCES Hotel(hotelno) );
```

Class exercise – Create tables

- Create table Students_xxxx (
id int,
name varchar(255),
zipcode varchar(12) ,
primary key (id));
- Create table Courses_xxxx (
id int,
name varchar(255),
s_id int ,
primary key (id, s_id),
foreign key (s_id) REFERENCES Students_xxxx(id));

See where the foreign key is referencing to

- Use the following SQL statement to know which table the foreign key is referencing to:
 - **Show create table the_TABLE_name;**
- **Note:** The copy statement won't copy P.K. and F.K.
 - **Create table myStaff as select * from dreamhome.Staff;**

MySQL errno 150

- Column name not matched between child and parent
- Table name not matched between child and parent.
(case sensitive)
- Column type or size not matched between child and parent.
- The reference column in the parent table is not a PK.

Referential Integrity & INSERT/UPDATE

- Any **INSERT/UPDATE** attempting to create FK value in child table without matching CK value in parent is rejected.
- Action taken attempting to **update/delete** a CK value in parent table with matching rows in child is dependent on referential action specified using ON UPDATE and ON DELETE subclauses:
 - CASCADE
 - SET DEFAULT
 - SET NULL
 - NO ACTION

Class exercise – Insert Records

- Insert into Students_xxxx (id, name, zipcode) values (1001, 'Austin', '07083'), (1002, 'Grant', '07021');
- Show all the records
- Insert records to Courses_xxxx table
Insert into Courses_xxxx values
(1, 'Java', 1001), (2, 'Database', 1001), (1, 'Java', 1002);
- Any error to run the following statement?
Insert into Courses_xxxx values
(5, 'Java', 1005);

Class exercise – drop table

- Can we drop Students_xxxx table ?
 - What is the problem?
 - ERROR 1217 (23000): Cannot delete or update a parent row: a foreign key constraint fails
 - How to drop it?
 - (1) drop foreign key first
 - (1.1) find foreign key constraint name (show create table)
show create table Courses_demo;
:
CONSTRAINT **Courses_demo_ibfk_1** FOREIGN KEY (`s\_id`)
REFERENCES `Students\_demo` (`id`)
:
(1.2) drop foreign key
alter table Courses_demo drop foreign key Courses_demo_ibfk_1;
 - (2) drop table
drop table Students_demo;

ALTER TABLE

- **Add a new column to a table.**
- **Drop a column from a table.**
- **Add a new table constraint.**
- **Drop a table constraint.**
- **Set a default for a column.**
- **Drop a default for a column.**

Example 7.2(a) - ALTER TABLE

Change Staff table by removing default of 'Assistant' for position column and setting default for sex column to female ('F').

ALTER TABLE Staff

ALTER position DROP DEFAULT;

ALTER TABLE Staff

ALTER sex SET DEFAULT 'F';

Example 7.2(b) - ALTER TABLE

Remove constraint from PropertyForRent that staff are not allowed to handle more than 100 properties at a time. Add new column to Client table.

ALTER TABLE PropertyForRent

DROP CONSTRAINT StaffNotHandlingTooMuch;

ALTER TABLE Client

ADD prefNoRooms PRooms;

Alter and auto increment

- Add an auto increment to an existing table
ALTER TABLE yourTableName **MODIFY** yourColumnName
INT NOT NULL **AUTO_INCREMENT**;
- Reset an auto increment
ALTER TABLE table_name AUTO_INCREMENT = **value**;
- <https://www.mysqltutorial.org/mysql-reset-auto-increment>

Add multiple columns in table

- The syntax to add multiple columns in a table in MySQL (using the ALTER TABLE statement) is:

ALTER TABLE table_name

ADD new_column_name column_definition

[**FIRST** | **AFTER** column_name],

ADD new_column_name column_definition

[**FIRST** | **AFTER** column_name],

...

;

- https://www.techonthenet.com/mysql/tables/alter_table.php

Example – add/modify multiple columns in a table

- Example – **add** multiple columns

ALTER TABLE contacts

ADD last_name varchar(40) NOT NULL

AFTER contact_id,

ADD first_name varchar(35) NULL

AFTER last_name;

- Example – **modify** multiple columns

ALTER TABLE table_name

MODIFY column_name column_definition

[**FIRST** | **AFTER** column_name],

MODIFY column_name column_definition

[**FIRST** | **AFTER** column_name],

...

;
https://www.techonthenet.com/mysql/tables/alter_table.php

Alter and change – column name and data type

- Change a column name in a table

ALTER TABLE your_table_name **RENAME COLUMN**
original_column_name **TO** new_column_name;

- Or

ALTER TABLE tablename

CHANGE oldname newname **data_type** ;

- We can also change the data type at the same if the data_type is different from the original one.
- <https://www.thoughtco.com/change-column-name-in-mysql-2693874>
- <https://phoenixnap.com/kb/how-to-rename-column-mysql>

Alter and change – column position

- The CHANGE also allows you to place the column in a different position in the table by using the optional **FIRST** | **AFTER** column_name clause.

- The new_col_name will be added after column_x.

ALTER TABLE your_table_name **CHANGE**
original_column_name new_col_name y_data_type
AFTER column_x;

- The new_col_name will be added at the first column.

ALTER TABLE your_table_name **CHANGE**
original_column_name new_col_name y_data_type **FIRST**;

Rename a table in MySQL

- The syntax to rename a table in MySQL is:

ALTER TABLE table_name

RENAME TO new_table_name;

https://www.techonthenet.com/mysql/tables/alter_table.php

- Or

RENAME TABLE old_table_name **TO** new_table_name;

- Note that you cannot use the RENAME TABLE statement to rename a temporary table, but you can use the ALTER TABLE statement to rename a temporary table.

- <https://www.mysqltutorial.org/mysql-rename-table/>

Class exercises 1

- https://www.w3schools.com/sql/sql_alter.asp
- xxxx is your Kean email ID

Create table Test_xxxx (id int, name varchar(10));

- Get permissions on the class website, option 6.3
- Insert two records

INSERT INTO Test_xxxx values

(1,'Test1'), (2,'Test2');

- Show records

Select * from Test_xxxxx;

Class exercises 2 (Alter)

- Add primary key constraint to id

```
ALTER TABLE Test_xxxx ADD PRIMARY KEY (id);
```

- Add auto_increment to id

```
ALTER TABLE Test_xxxx MODIFY COLUMN id INT auto_increment;
```

- Add NOT NULL constraint to name

```
ALTER TABLE Test_xxxx MODIFY column name varchar(10) NOT NULL;
```

- Add a new column zipcode, varchar(10)

```
ALTER TABLE Test_xxxx ADD column zipcode varchar(10);
```

- Add a foreign key constraint to zipcode reference to dreamhome.Zipcode(zipcode)

```
ALTER TABLE Test_xxxx ADD FOREIGN KEY (zipcode)  
REFERENCES dreamhome.Zipcode(zipcode);
```

Class exercises 3 (Alter)

- Add one column gender char(1) after id

Alter table Test_xxxx add gender char(1) after id;

- Show records

Select * from Test_xxxxx;

- Update the gender

Update Test_xxxx set gender='M' where id=1;

- Add primary key on id

Alter table Test_xxxx add primary key (id);

- Add column vid

Alter table Test_demo add column vid int first;

- Add foreign key

Alter table Test_xxxx add foreign key(vid) references dreamhome.Vendors(vid);

- Remove column gender

Alter table Test_xxxxx drop column gender;

Class exercises 4 (Alter) , part 1

```
mysql> select * from Test_demo;
```

```
+-----+-----+
| id    | name  |
+-----+-----+
| 1     | Test1 |
| 2     | Test2 |
+-----+-----+
```

- Change name to int

ALTER TABLE Test_xxxx modify name int;

→ What happen? **mysql> show warnings;**

```
mysql> ALTER TABLE Test_demo modify name int;
Query OK, 2 rows affected, 2 warnings (0.27 sec)
Records: 2  Duplicates: 0  Warnings: 2
```

```
mysql> show warnings;
```

```
+-----+-----+-----+
| Level   | Code | Message                                     |
+-----+-----+-----+
| Warning | 1366 | Incorrect integer value: 'Test1' for column 'name' at row 1 |
| Warning | 1366 | Incorrect integer value: 'Test2' for column 'name' at row 2 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> select * from Test_demo;
+-----+-----+
| id    | name |
+-----+-----+
| 1     | 0    |
| 2     | 0    |
+-----+-----+
```

Class exercises 4 (Alter) , part 2

- http://razorsql.com/features/mysql_rename_column.html
- Change column “name” to “myname” varchar(10)
`ALTER TABLE Test_xxxx change column `name` myname varchar(10);`
*** Dangerous: datatype change, data may be changed too.**
- Change table name
`RENAME TABLE Test_xxxx TO TTest_xxxx;`
- Can you drop TTest_xxxx? No/Yes
- Can you rename TTest_xxxx back to Test_xxxx? No/Yes

Class exercise – Create tables

- Create table Students_xxxx (
id int,
name varchar(255),
zipcode varchar(12) ,
primary key (id));

- Create table Courses_xxxx (
id int,
name varchar(255),
s_id int ,
primary key (id, s_id),
foreign key (s_id) REFERENCES Students_xxxx(id));

Class exercise – Insert Records 1

- Insert into Students_xxxx (id, name, zipcode) values (1001, 'Austin', '07083'), (1002, 'Grant', '07021');
- Show all the records
- Insert into Students_xxxx values (1003, 'Mary', '07083'), (1002, 'Sam', '07029');
 - What happens?

Class exercise – Insert Records 2

- Insert into Courses_xxxx values (3740, 'Database', 1001), (2231, 'Java', 1001);
- Show all the records
- Insert into Courses_xxxx values (3740, 'Database', 1005); ----What happens?
 - Fix the problem by changing 1005 to ?
 - Show all records.

Class exercise – Delete Records

- Delete from Students_xxxx where id= 1001;
- How to fix the problem?

Class exercise – Update Records

- Change id to 1100 fo Students_xxxx where id= 1001;
- How to fix the problem?

Class exercise – drop table

- Can we drop Students_xxxx table ?
 - What is the problem?
 - How to drop it?

Class exercise with Foreign key

- Create table Test (id int primary key, name varchar(10));
- Create table Test2 (id int primary key, name varchar(10), pid int, foreign key(pid) references Test(id));
- MySQL errno: 150 → F.K. error
 - Parent table/column name, F.K. data type/size
- Insert order – parent first
 - Insert into Test2 values (1,'Austin',1); --what happen?
 - Insert into Test values(1,'Monkey');

DELETE, UPDATE with Foreign key

- Delete order – child first
 - Delete from Test_xxxx where id=1; -- what happen?
 - Delete from Test2_xxxx where pid = 1;
- Insert the data again.
- How about update?
 - Update Test_xxxx set id=2; -- what happen
 - SET foreign_key_checks = 0;
 - Update Test_xxxx set id=2 where id=1;
 - Update Test2_xxxx set pid=2 where pid=1;
 - SET foreign_key_checks = 1;
- Too much trouble if we need to update many records with FK. → Cascade

CASCADE in the Foreign Key

- CASCADE: Delete or update the row from the parent table, and automatically delete or update the matching rows in the child table.
- Both ON DELETE CASCADE and ON UPDATE CASCADE are supported.

FOREIGN KEY (*col_name*)

REFERENCES *tbl_name* (*index_col_name*)

[ON DELETE CASCADE]

[ON UPDATE CASCADE]

<https://dev.mysql.com/doc/refman/5.7/en/create-table-foreign-keys.html>

Example of Referential Integrity & CASCADE

- The cascade should be set on the child table because only the Foreign Key is defined in the child table.

Create table Test2_xxxx (id int primary key, name varchar(10), pid int, **foreign key(pid) references Test_demo(id) ON DELETE SET NULL**);

Create table Test2_xxxx (id int primary key, name varchar(10), pid int, **foreign key(pid) references Test_demo(id) ON UPDATE CASCADE**);

CASCADE - CREATE TABLES

- Create the following two tables

```
CREATE TABLE Test_demo(  
    id INT PRIMARY KEY,  
    name VARCHAR(10) NOT NULL);
```

```
CREATE TABLE Test3_demo(  
    uid INT PRIMARY KEY,  
    tid INT NOT NULL,  
    FOREIGN KEY (tid) REFERENCES Test_demo(id)  
    ON DELETE CASCADE  
    ON UPDATE CASCADE);
```

CASCADE - DELETE

- INSERT 2-3 records into each table

INSERT INTO Test_demo values (1,'Test1'),(2,'Test2');

INSERT INTO Test3_demo values (1,1),(2,1),(3,2);

- What happen if we delete id=2 in Test_demo table

DELETE FROM Test_demo where id=2;

- How many records in Test3_demo table?
- Can we delete the record if there is no cascade clause when we created Test_demo table?

Home exercises

- Work on MySQL workbench, xxxx is your DB login.
- Create a table Test1_xxxx with columns
 - branchno, name
 - branchno is the primary key
- Create a table Test2_xxxx with columns
 - id, name, birthday, gender, salary, bno
 - Assign proper data type for each column
 - Primary key : id
 - Foreign key: bno references to Test1_xxxx(branchno)
ON DELETE CASCADE
- Insert 3 records into each table
- Delete 1 record from Test1_xxxx
- Change the bno in Test2_xxxx
- Drop Test2_xxxx table

Reading Assignment & Exercises

- Create three tables
 - **Students_xxxx(sid, name, major)**
 - **Courses_xxxx(cid, name, credits)**
 - **Students_Courses_xxxx(sid, cid, semester, year, grade)**
where xxxx is your email ID
- Insert 3 records into each table
- Add **sid** as **PK** in Student and **FK** in Students_Courses
- Add a new column birthday in the student table
- Update values
- Delete records
- Remove and add foreign key

Conclusion

- Work on project
- CREATE with Foreign key
 - Cascade
- Do all exercises in Chapter 7